

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	60310	52.314739
1	[Git, Change, Create]	18094	15.695289
2	[Git, Commit]	11441	9.924273
3	[Git, Change, Delete]	8846	7.673291
4	[File, Git]	5953	5.163814
5	[Git, Change, Rename]	3403	2.951866
6	[Git, Tag]	1843	1.598675
7	[Author, Git, Person]	1272	1.103372
8	[Json, Key]	668	0.579444
9	[Json, Value, Scalar]	603	0.523061
10	[Committer, Git, Person]	370	0.320949
11	[NPM, Dependency]	338	0.293192
12	[Type, TS, Primitive]	293	0.254157
13	[Type, TS, Declared]	277	0.240278
14	[TS, ExternalDeclaration]	215	0.186498
15	[Git, Change, Copy]	156	0.135319
16	[Type, TS, Literal]	136	0.117971
17	[Json, Value, Object]	133	0.115368
18	[Type, TS, Union]	120	0.104092
19	[Type, TS, ObjectMember]	103	0.089345
20	[NPM, Script]	91	0.078936
21	[TS, Property]	65	0.056383
22	[TS, Function]	47	0.040769
23	[Type, TS, FunctionParameter]	40	0.034697
24	[Type, Object, TS]	39	0.033830
25	[Git, Branch]	37	0.032095
26	[File, Directory]	34	0.029493
27	[Type, TS, Function]	34	0.029493
28	[TS, Parameter]	33	0.028625
29	[Package, File, Json, NPM]	29	0.025155

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

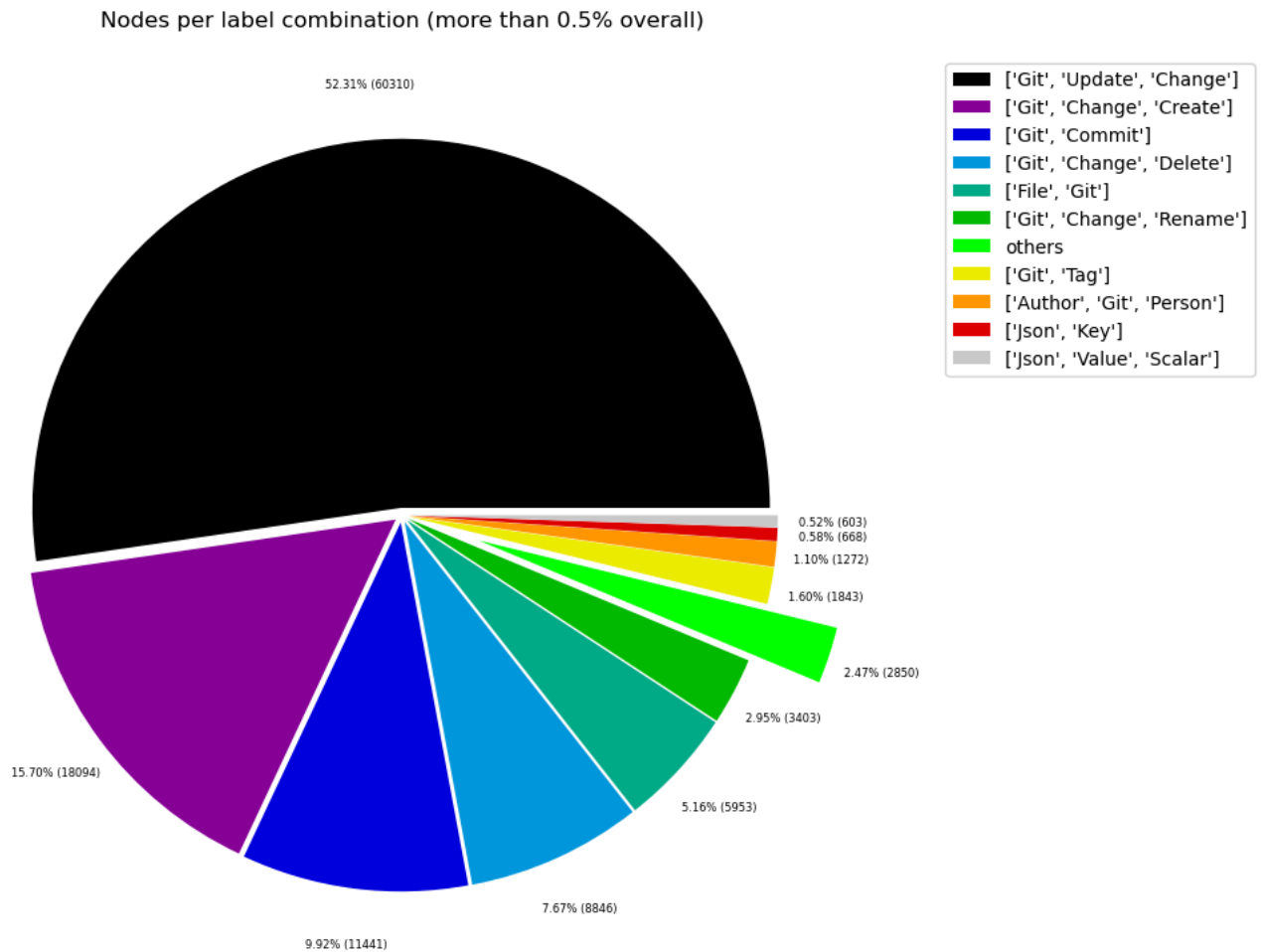


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000867
1	[NPM, PublishConfig]	1	0.000867
2	[File, TS, Scan]	1	0.000867
3	[TS, Method]	1	0.000867
4	[Repository, File, Git]	1	0.000867
5	[TS, Constructor]	1	0.000867
6	[Value, TS, ObjectMember]	1	0.000867
7	[TS, Class]	1	0.000867
8	[TS, Enum]	2	0.001735
9	[Value, Object, TS]	3	0.002602
10	[Type, TS, Tuple]	3	0.002602
11	[Value, TS, Function]	4	0.003470
12	[TS, TypeParameter]	4	0.003470
13	[Value, TS, Complex]	5	0.004337
14	[Repository, NPM]	5	0.004337
15	[NPM, Engine]	6	0.005205
16	[Project, TS]	6	0.005205
17	[File, Local]	6	0.005205
18	[Type, TS, TypeParameterReference]	6	0.005205
19	[Value, TS, Member]	6	0.005205
20	[File, TS, Local, Module]	6	0.005205
21	[Value, TS, Call]	6	0.005205
22	[TS, EnumMember]	8	0.006939
23	[Type, TS, NotIdentified]	11	0.009542
24	[TS, ExternalModule]	11	0.009542
25	[Json, Value, Array]	12	0.010409
26	[Value, TS, Declared]	13	0.011277
27	[TS, TypeAlias]	16	0.013879
28	[File, Directory, Local]	16	0.013879
29	[TS, Interface]	17	0.014746

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

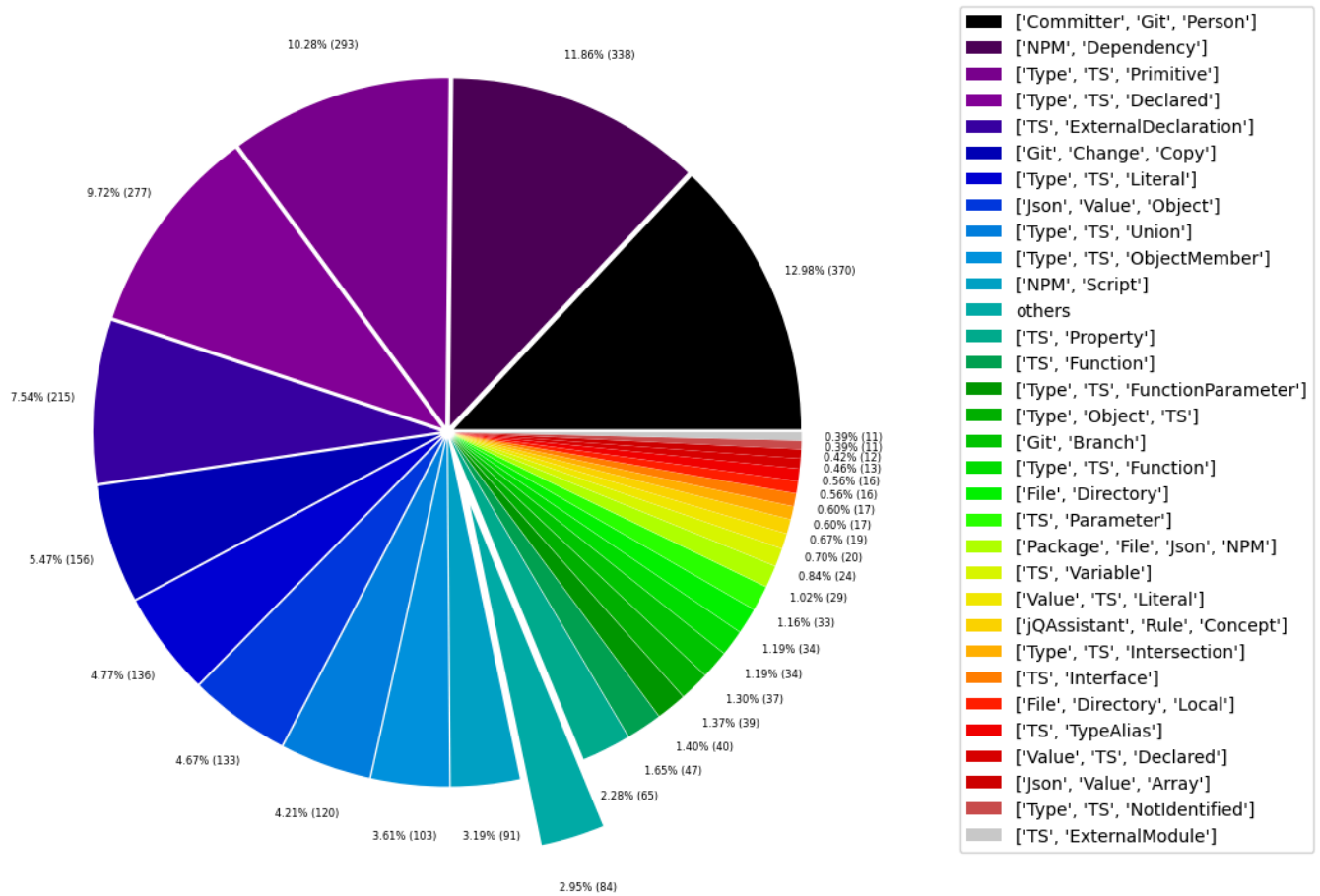


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

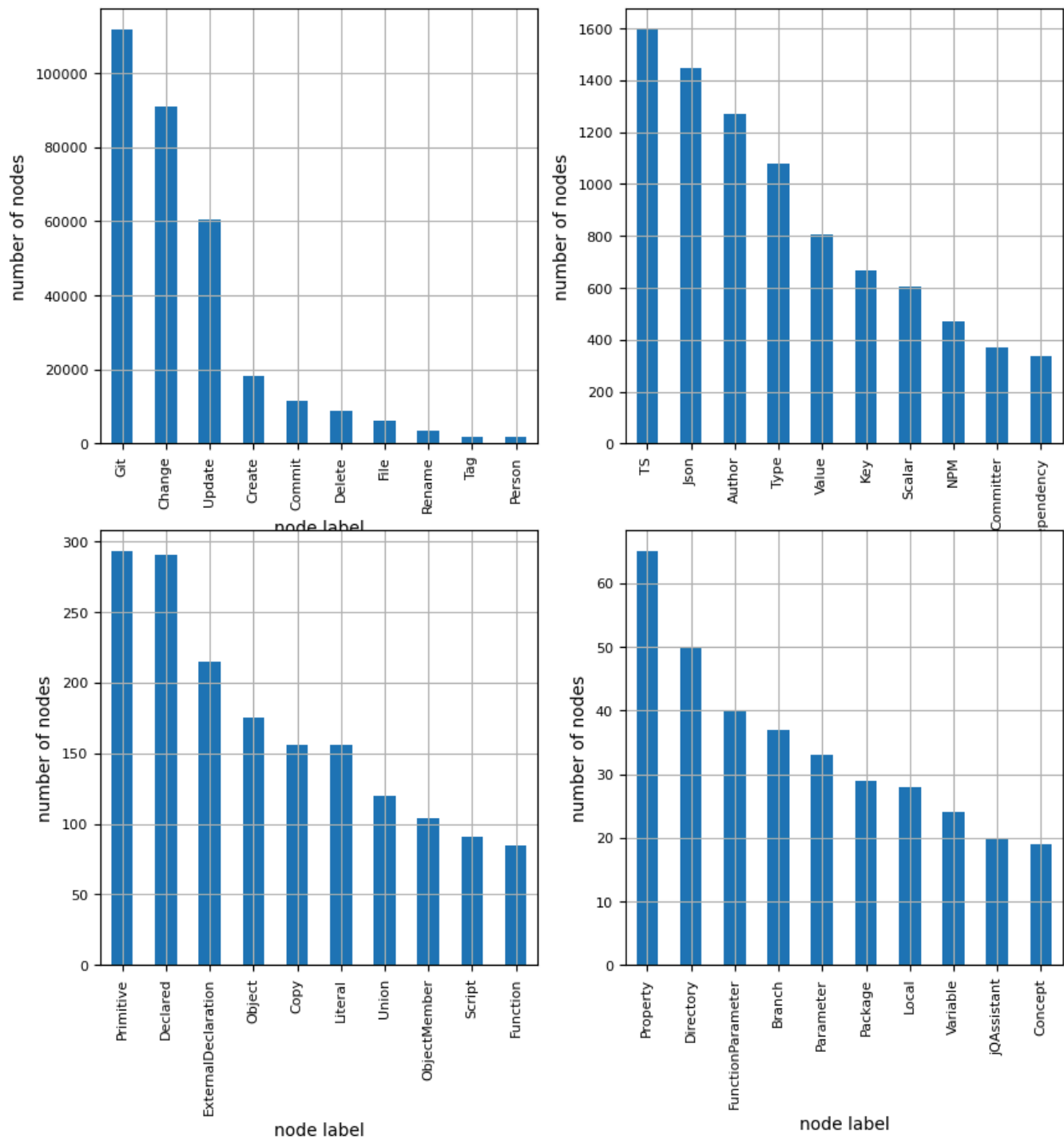
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	111726	96.914549
1	Change	90809	78.770504
2	Update	60310	52.314739
3	Create	18094	15.695289
4	Commit	11441	9.924273
5	Delete	8846	7.673291
6	File	6046	5.244485
7	Rename	3403	2.951866
8	Tag	1843	1.598675
9	Person	1642	1.424321
10	TS	1595	1.383552
11	Json	1445	1.253437
12	Author	1272	1.103372
13	Type	1079	0.935958
14	Value	806	0.699149
15	Key	668	0.579444
16	Scalar	603	0.523061
17	NPM	470	0.407692
18	Committer	370	0.320949
19	Dependency	338	0.293192
20	Primitive	293	0.254157
21	Declared	290	0.251555
22	ExternalDeclaration	215	0.186498
23	Object	175	0.151800
24	Copy	156	0.135319
25	Literal	156	0.135319
26	Union	120	0.104092
27	ObjectMember	104	0.090213
28	Script	91	0.078936
29	Function	85	0.073732
30	Property	65	0.056383
31	Directory	50	0.043372
32	FunctionParameter	40	0.034697
33	Branch	37	0.032095
34	Parameter	33	0.028625
35	Package	29	0.025155
36	Local	28	0.024288
37	Variable	24	0.020818
38	jqAssistant	20	0.017349
39	Concept	19	0.016481

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 346317

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	90809	26.221352
1	MODIFIES	90809	26.221352
2	UPDATES	60310	17.414681
3	COMMITTED	22882	6.607241
4	CREATES	21653	6.252364
5	HAS_PARENT	12540	3.620960
6	DELETES	12249	3.536933
7	HAS_COMMIT	11441	3.303621
8	HAS_FILE	5953	1.718945
9	RENAMES	3403	0.982626
10	CHANGED_TOGETHER_WITH	1931	0.557582
11	HAS_TAG	1843	0.532171
12	ON_COMMIT	1843	0.532171
13	HAS_NEW_NAME	1784	0.515135
14	HAS_AUTHOR	1272	0.367294
15	DEPENDS_ON	887	0.256124
16	HAS_KEY	668	0.192887
17	HAS_VALUE	668	0.192887
18	CONTAINS	596	0.172097
19	HAS_COMMITTER	370	0.106839
20	OF_TYPE	339	0.097887
21	EXPORTS	309	0.089225
22	REFERENCES	197	0.056884
23	DECLARES	186	0.053708
24	DECLARES_DEV_DEPENDENCY	169	0.048799
25	DECLARES_DEPENDENCY	161	0.046489
26	COPIES	156	0.045045
27	HAS_MEMBER	104	0.030030
28	HAS_TYPE_ARGUMENT	94	0.027143
29	DECLARES_SCRIPT	91	0.026277

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

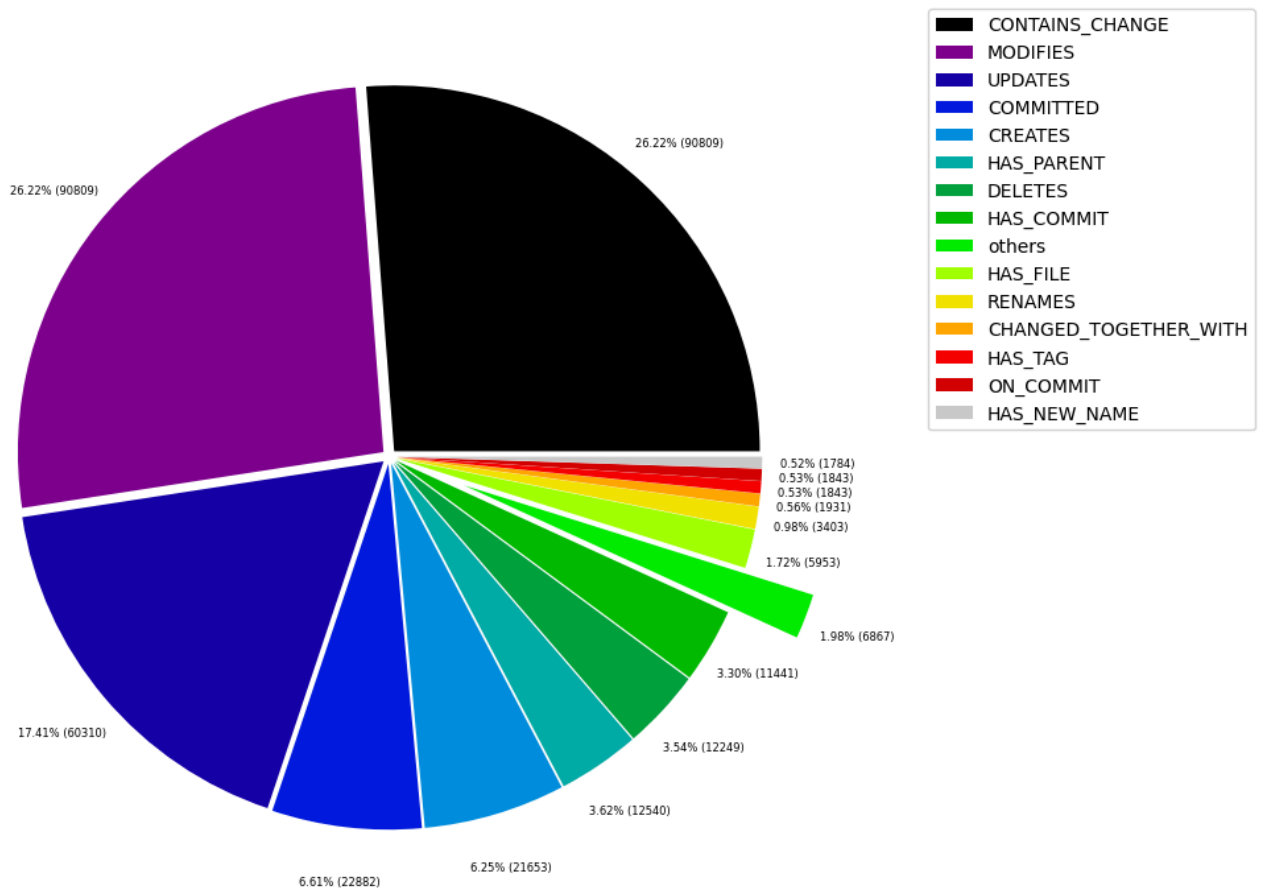


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

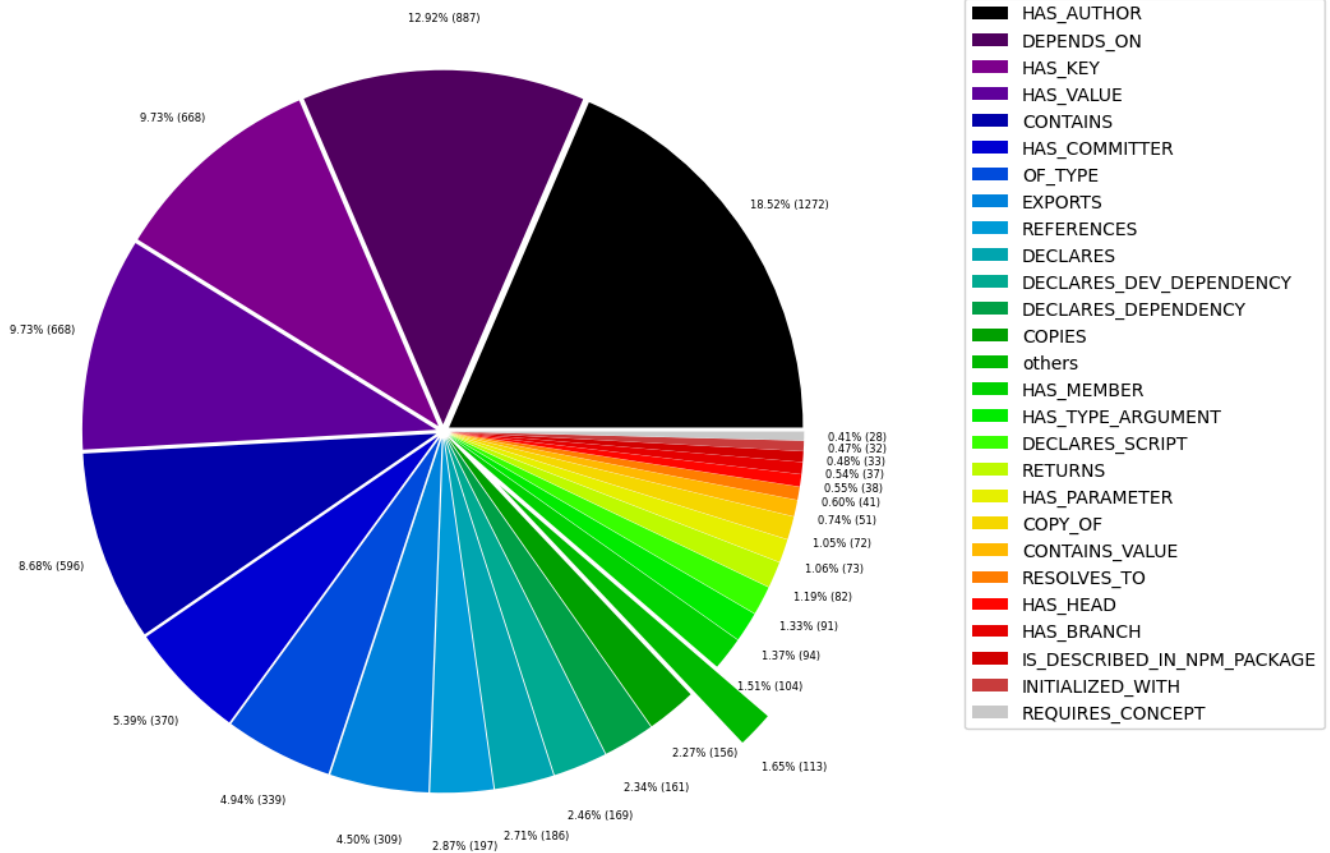
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000289
1	CONSTRAINED_BY	4	0.001155
2	IN_REPOSITORY	5	0.001444
3	REFERENCED_PROJECTS	5	0.001444
4	PARENT	6	0.001733
5	MEMBER	6	0.001733
6	HAS_ROOT	6	0.001733
7	HAS_NPM_PACKAGE	6	0.001733
8	HAS_CONFIG	6	0.001733
9	HAS_ARGUMENT	6	0.001733
10	EXTENDS	6	0.001733
11	DECLARES_ENGINE	6	0.001733
12	CONTAINS_PROJECT	6	0.001733
13	CALLS	6	0.001733
14	DECLARES_PEER_DEPENDENCY	8	0.002310
15	USES	11	0.003176
16	INCLUDES_CONCEPT	19	0.005486
17	REQUIRES_CONCEPT	28	0.008085
18	INITIALIZED_WITH	32	0.009240
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009529
20	HAS_BRANCH	37	0.010684
21	HAS_HEAD	38	0.010973
22	RESOLVES_TO	41	0.011839
23	CONTAINS_VALUE	51	0.014726
24	COPY_OF	72	0.020790
25	HAS_PARAMETER	73	0.021079
26	RETURNS	82	0.023678
27	DECLARES_SCRIPT	91	0.026277
28	HAS_TYPE_ARGUMENT	94	0.027143
29	HAS_MEMBER	104	0.030030

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodesWithSameLabelsAsTarget
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	60310	11441	
1	[Git, Update, Change]	MODIFIES	[File, Git]	60310	60310	
2	[Git, Update, Change]	UPDATES	[File, Git]	60310	60310	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	18094	11441	
4	[Git, Change, Create]	CREATES	[File, Git]	18094	18094	
5	[Git, Change, Create]	MODIFIES	[File, Git]	18094	18094	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12540	11441	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11441	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11441	1272	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11441	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8846	11441	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8846	8846	
12	[Git, Change, Delete]	DELETES	[File, Git]	8846	8846	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5953	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3403	11441	
15	[Git, Change, Rename]	DELETES	[File, Git]	3403	3403	
16	[Git, Change, Rename]	CREATES	[File, Git]	3403	3403	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3403	3403	
18	[Git, Change, Rename]	RENAMES	[File, Git]	3403	3403	
19	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1843	1	
20	[Git, Tag]	ON_COMMIT	[Git, Commit]	1843	1843	
21	[File, Git]	HAS_NEW_NAME	[File, Git]	1784	5953	
22	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1676	5953	
23	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1272	1	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[Package, File, Json, NPM]	CHANGED_TOGETHER_WITH	[Package, File, Json, NPM]	253	29	
29	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	

Graph Density

total_number_of_nodes (vertices): 115283

total_number_of_relationships (edges): 346317

-> total directed graph density: 2.6058357550463212e-05

-> total directed graph density in percent: 0.0026058357550463214